Many Treatments, Some Discrete Instruments:

Choice Models and the Identification of Causal Effects

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IV with Multiple Treatments

Binary treatment: under a "monotonicity" assumption, IV estimand identifies LATE (Imbens and Angrist, 1994)

- in some cases, LATE is policy relevant
- in other cases, different parameters are of interest (external validity)

Multiple treatments: point identification can be hard in practice.

In this case, assumptions common in the binary setting are often

- implausible in multiple treatment settings
- no longer sufficient for point-identification of relevant parameters
- too restrictive for many real-world applications

IV with Multiple Treatments

Question

What can we learn about policy relevant parameters that are not point-identified by the exogenous variation we observe?

Goal

Inference for many different choices of parameters, under many different choices of assumptions on treatment selection

This paper's focus

- Discrete-valued treatment $\rightarrow$ may be ordered or unordered
- Discrete-valued instruments: $\rightarrow$ common case in empirical work
- Randomly assigned instruments $\rightarrow$ not just mean independent

Basic Model

Outcome

Treatment $D \in \mathcal{D} := \{0, \ldots, N_D\}$, (counterfactual) potential outcomes $\left(Y_0, \ldots, Y_{N_D}\right)$, and actual (realized) outcome

$$Y = \sum_{d \in \mathcal{D}} Y_d \mathbb{1}\{D = d\} .$$

Denote by $\mathcal{Y}$ the support of Y.

If $\mathcal{Y}$ is not finite, we require that $\mathcal{Y}$ is a compact convex subset of $\mathbb{R}^n$
$\to$ outcome Y can be continuous

Today: For simplicity, assume Y is continuous and $\mathcal{Y} \subseteq \mathbb{R}$ is a compact interval

Basic Model

Selection into Treatments

Instrument $Z \in \mathcal{Z} := \{0, \dots, N_Z\}$, potential treatments $\left(D_0, \dots, D_{N_Z}\right)$, and actual treatment

$$D = \sum_{z \in \mathcal{Z}} D_z \mathbb{1}\{Z = z\} .$$

Instrument Exogeneity Assumption

Require that Z is randomly assigned

$$\left(Y_0, \dots, Y_{N_D}, D_0, \dots, D_{N_Z}\right) \perp Z .$$

We do not *require* any additional assumptions, but can *incorporate* additional assumptions on relationship between Z and D.

Basic Model

Comments

- Covariates omitted to simplify exposition
  - All results and assumptions can be viewed as holding conditional on any observed baseline covariates

- **Generally:** Framework easily accomodates settings where Y, D, and/or Z are only partially observed
  - Examples: missing wages for unemployed, imprecisely observed treatment, modelling incentives not seen in data etc.

- **Today:** For simplicity, assume that (Y, D, Z) is observed

Structural IV Model

The basic model above is equivalent to the following model, with observables (Y, D, Z) as before, and unobservables $(\mathbf{V}, \varepsilon_Y)$:

$$\text{Treatment Choice Equation:} \quad D = f_D(Z, \mathbf{V}) \ , \tag{1}$$

$$\text{Outcome Equation:} \quad Y = f_Y\left(D, \mathbf{V}, \varepsilon_Y\right) \ , \tag{2}$$

$$\text{Independence Condition:} \quad Z, \mathbf{V}, \varepsilon_Y \text{ are mutually independent} \tag{3}$$

- $\mathbf{V}$ is an unobserved confounder that can generate selection bias
- ε_Y is an unobserved error term
- $\mathbf{V}$ and ε_Y can be infinite-dimensional (unrestricted heterogeneity)

$\rightarrow$ **We allow:** assumptions on treatment selection, i.e. on f_D and/or $\mathbf{V}$

$\rightarrow$ **We avoid:** assumptions on outcome equation, i.e. on f_Y or ε_Y

Structural IV Model

Comments

- Potential outcomes and treatments can be defined by

$$Y_d := f_Y\left(d, \mathbf{V}, \varepsilon_Y\right) \text{ and } D_z := f_D(z, \mathbf{V})$$

- $Z, \mathbf{V}$, and ε_Y are mutually independent $\implies$ Z is randomly assigned
- We can allow some types of assumptions on outcomes: assumptions like $E\left(Y_d\right) \geq E\left(Y_{d'}\right)$ can be easily incorporated, but generally not assumptions like $P\left(Y_d \geq Y_{d'}\right) = 1$... (more on this later)
- Assumptions on f_D and/or $\mathbf{V}$ provide identifying power in our model only through restrictions on the (unobserved) joint distribution of $\left(D_0, \ldots, D_{N_Z}\right)$

Response-Types

Define an individual's ***response-type***, denoted T, as the (random) vector of their potential treatments[1]

$$T(\mathbf{V}) = T := \left(D_0, \dots, D_{N_Z}\right) = (f_D(0, \mathbf{V}), \dots, f_D(N_Z, \mathbf{V}))$$

Define $\mathbf{T} := \text{supp}(T) = \left\{\tau_0, \dots, \tau_{N_T}\right\}$ as the support of T (i.e. possible values for T). Note that

- $\mathbf{T} = \{T(v) : v \in \text{supp}(\mathbf{V})\} = \{(f_D(0, v), \dots, f_D(N_Z, v)) : v \in \text{supp}(\mathbf{V})\}$
- $\mathbf{T}$ depends on the assumptions we impose
- If we impose no additional assumptions, then $\mathbf{T} = \mathcal{D}^{|\mathcal{Z}|}$

Notation: For any response-type $\tau \in \mathcal{D}^{|\mathcal{Z}|}$, denote by $\tau(z)$ the treatment taken by individuals of response-type $T = \tau$ when $Z = z$.

[1] This notion of response-type is the same as Heckman and Pinto (2018); often referred to as "principal strata" in the statistics literature.

Response-Types

Basic Example: Suppose $Z \in \{0,1\}$ and $D \in \{0,1\}$ are binary. Without additional assumptions, $\mathbf{T}$ consists of the following response-types, (D_0, D_1):

$$\underbrace{(1,1)}_{\text{always-takers}} \qquad \underbrace{(0,1)}_{\text{compliers}} \qquad \underbrace{(1,0)}_{\text{defiers}} \qquad \underbrace{(0,0)}_{\text{never-takers}}$$

(or 1-always-takers) (or 0-always-takers)

In this case, adding the "monotonicity" assumption (e.g., by requiring that $f_D(\cdot, \mathbf{V})$ is an increasing function), eliminates the "defiers" type.

Under monotonicity, $\mathbf{T} = \left\{\tau \in \{0,1\}^2 : \tau(0) \leq \tau(1)\right\}$.

Response-Types

Comments

- Confounder $\mathbf{V}$ governs how individuals select into treatment
- Conditional on T, treatment only depends on Z, which is randomly assigned (and therefore independent of $\mathbf{V}$)
- Response-types define a coarse partition of the support of $\mathbf{V}$:

$$\{v \in T^{-1}(\tau)\} = \{v \in \operatorname{supp}(\mathbf{V}) : T(v) = \tau\}$$

- We can define a generalized Marginal Treatment Response (MTR) function based on $\mathbf{V}$,

$$MTR_d(v) := E(Y_d \mid \mathbf{V} = v) \ ,$$

  and get that $E(Y_d \mid T = \tau) = \int_{T^{-1}(\tau)} MTR_d(v) \ d\mu(v)$

Primary Parameters of Interest

We assume that the parameter of interest, β, is of the form

$$\beta = \frac{\sum_{i \in \mathcal{I}_\beta} \omega_i P(T = \tau_i) E\left(Y_{d_i} \mid T = \tau_i\right)}{\sum_{j \in \mathcal{I}_\beta} \omega_j P\left(T = \tau_j\right)},$$

for known (or identified) values $\left\{(\omega_i, d_i, \tau_i) : i \in I_\beta\right\} \subseteq \mathbb{R} \times \mathcal{D} \times \mathbf{T}$.

Includes all parameters that are some weighted average of "local" average treatment effects, i.e.

$$\beta = E\left(Y_d - Y_{d'} \mid T \in \mathbf{T}_\beta\right) = \frac{\sum_{\tau \in \mathbf{T}_\beta} P(T = \tau) E(Y_d - Y_{d'} \mid T = \tau)}{\sum_{\tau' \in \mathbf{T}_\beta} P(T = \tau')},$$

for some subset of types $\mathbf{T}_\beta \subseteq \mathbf{T}$.

Examples: ATE, ATT, ATUT, (generalized versions of) LATE, PRTE etc.

Modelling Treatment Selection: Choice Restrictions

Assumptions on f_D and/or $\mathbf{V}$ provide identifying power by restricting the distribution (and support) of T. We call such assumptions ***choice restrictions***.

Generally: Allow all choice restrictions that restrict the distribution of T to lie in a convex subset of the set of all probability mass functions on $\mathcal{D}^{\mathcal{Z}}$ (i.e. the $\mathcal{D}^{\mathcal{Z}}$-simplex, $\Delta^{\mathcal{D}^{\mathcal{Z}}}$).

Today: Focus on choice restrictions that simply restrict $\mathbf{T}$, the support of T, (i.e. assumptions that simply eliminate some types). We will refer to such assumptions as ***type restrictions***.

Example: Job Training and Wages with Firm Heterogeneity

(work in progress with Kory Kroft and Ismael Mourifié)

Evaluating impact of randomly assigned job training program on wages can be challenging:

- Wages are only observed for those who are employed (sample selection)
- Employment status may itself be affected by the training program (sample selection bias)

Let $D \in \{0, 1\}$ be employment status, and let $Z \in \{0, 1\}$ represent randomly assigned job training. Assume $D_1 \geq D_0$.

$\rightarrow$ Lee (2009) seeks to estimate "human capital" effect of job training, proposes the partially identified parameter:

$$E\left(Y_1 - Y_0 \mid D_1 = D_0 = 1\right) = E\left(Y_1 - Y_0 \mid T = \text{always-employed}\right)$$

Example: Job Training and Wages with Firm Heterogeneity

Suppose there are two firms:
firm L ("low-productivity") and firm H ("high-productivity")

- Now $D \in \{0, L, H\}$ represents employment status, including which firm you are employed at (0 is still unemployed)
- Let $J = Z$ be job-training status, treatment is now the tuple (J, D)
- Outcomes depend on both training status and firm-type, e.g.
  - $Y_{1,L}$ is wages for worker if they were trained and employed at firm L
  - $Y_{0,L}$ is wages for worker if they were untrained and employed at firm L
- Response-types are indexed by (D_0, D_1), since $J = Z$ for all response-types
  - e.g. (L, H) is a type of "always-employed" that switches employment from firm L to H due to training

Example: Job Training and Wages with Firm Heterogeneity

In this setting $E\left(Y_{1,\cdot} - Y_{0,\cdot} \mid T = \text{always-employed}\right)$ captures the following weighted average of effects:

$$\left.\begin{aligned} &\omega_1 E\left(Y_{1,L} - Y_{0,L} \mid T \in \{(L,L),(L,H)\}\right) \\ &+ \omega_2 E\left(Y_{1,H} - Y_{0,H} \mid T \in \{(H,H),(H,L)\}\right) \end{aligned}\right\} \text{“human capital” effects}$$

$$\left.\begin{aligned} &+ \omega_3 E\left(Y_{1,H} - Y_{1,L} \mid T \in \{(L,H)\}\right) \\ &+ \omega_4 E\left(Y_{1,L} - Y_{1,H} \mid T \in \{(H,L)\}\right) \end{aligned}\right\} \text{“sorting” effects}$$

where the ω_i are non-negative weights.

We may be interested in disentangling these two channels. For example, β can be a firm-specific ATE like $E\left(Y_{1,H} - Y_{0,H}\right)$, or a local effect like

$$E\left(Y_{1,H} - Y_{0,H} \mid T = (H,H)\right) .$$

More Examples of Potential Applications

- Multi-dimensional instrument(s)
  - More generally: settings with complex incentive schemes, or complicated relationship between Z and D
- Multi-dimensional treatment: disentangling different dimensions
  - Mediation: want to separate direct, indirect, and total effects
- Ordered Treatments: e.g., Impact of Incarceration, Schooling etc.
- (Nonparametric) Discrete Choice Models: No College vs 2-Year vs 4-Year College choice, Neighbourhood Choice etc.
  - MTO Experiment: see papers by Chetty, Katz, Pinto, and co-authors...
- Exclusion Restriction Violations and/or Sample Selection Problems
- Other settings with multi-dimensional heterogeneity
  - Survey/RCT Non-Response Bias (e.g., Dutz et al., 2022)
- Sensitivity analysis for point-identified models
  - In the spirit of Noack (2021)

Related IV Literature

Point Identification: Imbens and Angrist (1994)... Heckman and Vytlacil (1999, 2001, 2007a,b)... Heckman and Pinto (2018); Lee and Salanié (2018); Mountjoy (2022); Heinesen et al. (2022)... (and many more!)

Partial Identification: Manski (1990); Manski and Pepper (2000, 2009); Huber et al. (2017); Mogstad et al. (2018); Torgovitsky (2019); Mourifié et al. (2020); Lee and Salanié (2023); Mogstad et al. (2023); Marx (2023); Kamat et al. (2023); Gu and Russell (2023); Han and Yang (2023)...

→ **This paper:** provides sharp bounds; requires randomly assigned discrete instruments; incorporates assumptions on treatment selection; allows continuous outcomes; no (semi-)parametric assumptions required

→ **This paper cannot:** incorporate continuous instruments, mean-independent instruments, or stronger assumptions on outcomes (e.g. MTE shape restrictions, classical Roy model, rank similarity etc.)

Other Related Literature

Specification Tests, Sensitivity Analysis: Kitagawa (2015); Mourifié and Wan (2017); Kédagni and Mourifié (2020); Noack (2021); Sun (2023)...

→ **This paper:** sharp test for a broad class of choice restrictions

Joint Distribution of Potential Outcomes: Heckman and Smith (1995, 1998); Heckman et al. (1997); Jun and Lee (2023); Kaji (2023)...

Nonparametric Mixture and Sample Selection Models: Horowitz and Manski (1995, 1998, 2000); Lee (2009); Henry et al. (2014)... (many more)

Inference on Convex Identified Sets: Beresteanu and Molinari (2008) Beresteanu et al. (2011); Chernozhukov et al. (2013); Andrews and Shi (2013); Fang and Seo (2021); Fang et al. (2023); Cho and Russell (2023)...

Our problems: polyhedral cone inclusion problem (for specification test) and special class of convex programs (for inference on parameters)

→ **Goal:** uniformly valid and nonconservative inference (in progress...)

Plan of Talk

1. Informal Overview of Approach
2. Preliminaries and Formal Setup
3. Main Identification Results
4. Basic Outline of Inference Strategy
5. Conclusion

Basic Idea: IV Mixture Model

For any $d \in \mathcal{D}$ and each $z \in \mathcal{Z}$,

→ **We observe:** Y for those assigned $Z = z$ and took treatment d

→ **We want to learn about:** Y_d for those who are of type $T = \tau$, for some $\tau \in \mathbf{T}$

Key: The collection of individuals who take treatment $D = d$ when $Z = z$ consists of all individuals that are of types $T = \tau$ such that $\tau(z) = d$.

Basic Idea: IV Mixture Model

Observed outcome distributions are mixture distributions:

$$
\begin{aligned}
P(Y \leq y \mid D = d, Z = z) &= P(Y_d \leq y \mid D_z = d, Z = z) \\
&= P(Y_d \leq y \mid D_z = d) \\
&= \sum_{\tau \in \mathbf{T} : \tau(z) = d} \left(\frac{P(T = \tau)}{P(D = d \mid Z = z)} \right) P(Y_d \leq y \mid T = \tau)
\end{aligned}
$$

In terms of densities,

$$
\begin{aligned}
\underbrace{P(D = d \mid Z = z) f_{Y \mid D = d, Z = z}(y)}_{\text{observed}} &= \sum_{\tau \in \mathbf{T} : \tau(z) = d} P(T = \tau) f_{Y_d \mid T = \tau}(y) \\
&= \sum_{\tau \in \mathbf{T}} \underbrace{\mathbb{1}\{\tau(z) = d\}}_{\substack{\text{known,} \\ \text{from model}}} \underbrace{P(T = \tau) f_{Y_d \mid T = \tau}(y)}_{\substack{\text{unobserved,} \\ \text{primitive}}} .
\end{aligned}
$$

Basic Idea: IV Mixture Model

For every treatment $d \in \mathcal{D}$ and outcome-level $y \in \mathcal{Y}$, we have the following linear system with known coefficients

$$\sum_{\tau \in \mathbf{T}} \mathbb{1}\{\tau(0) = d\} \left(P(T = \tau) f_{Y_d|T=\tau}(y) \right) = P(D = d|Z = 0) f_{Y|D=d,Z=0}(y)$$

$$\vdots \qquad (\star)$$

$$\sum_{\tau \in \mathbf{T}} \mathbb{1}\{\tau(N_Z) = d\} \left(P(T = \tau) f_{Y_d|T=\tau}(y) \right) = P(D = d|Z = N_Z) f_{Y|D=d,Z=N_Z}(y)$$

$\rightarrow$ For each $y \in \mathcal{Y}$, the (unobserved) vector $\left[P(T = \tau) f_{Y_d|T=\tau}(y) : \tau \in \mathbf{T} \right]$ must be a solution to $(\star)$

$\rightarrow$ This restricts the weighted density function $P(T = \tau) f_{Y_d|T=\tau}(\cdot)$

Idea: "Integrate restrictions" over $y \in \mathcal{Y}$ to get restrictions on $P(T = \tau)$ and $P(T = \tau) E(Y_d \mid T = \tau)$

Notation

For $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, define $\mathbf{pf}_{Y|d,Z}$ as the observed vector of weighted densities

$$\mathbf{pf}_{Y|d,Z}(y) := \begin{bmatrix} P(D=d|Z=0)\, f_{Y|D=d,Z=0}(y) \\ \vdots \\ P(D=d|Z=N_Z)\, f_{Y|D=d,Z=N_Z}(y) \end{bmatrix},$$

and define $\boldsymbol{\pi\rho}_{Y_d|T}$ as the unobserved vector of weighted response-type densities

$$\boldsymbol{\pi\rho}_{Y_d|T}(y) := \begin{bmatrix} P(T=\tau_0)\, f_{Y_d|T=\tau_0}(y) \\ \vdots \\ P\left(T=\tau_{N_T}\right) f_{Y_d|T=\tau_{N_T}}(y) \end{bmatrix}.$$

Also define $\boldsymbol{\pi}$ as the unobserved vector of response-type probabilities,

$$\boldsymbol{\pi} := \begin{bmatrix} P(T=\tau_0) \\ \vdots \\ P\left(T=\tau_{N_T}\right) \end{bmatrix}.$$

Notation

Finally, collect the coefficients in ($\star$) into the following known matrix

$$\mathbf{R}_d := \begin{bmatrix} \mathbb{1}\{\tau_0(0) = d\} & \dots & \mathbb{1}\left\{\tau_{N_T}(0) = d\right\} \\ \vdots & \ddots & \vdots \\ \mathbb{1}\{\tau_0(N_Z) = d\} & \dots & \mathbb{1}\left\{\tau_{N_T}(N_Z) = d\right\} \end{bmatrix},$$

which indicates when each response-type takes treatment d (as in Heckman and Pinto, 2018).

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, no choice restrictions) :

In this case, the response types are always-takers (AT), compliers (C), defiers (D), never-takers (NT) , leading to

$$\mathbf{R}_0 = \begin{array}{c} \begin{array}{cccc} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array} \begin{array}{c} \\ (Z=0) \\ (Z=1) \end{array} \quad \text{and} \quad \mathbf{R}_1 = \begin{array}{c} \begin{array}{cccc} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{c} \\ (Z=0) \\ (Z=1) \end{array}$$

Notation

Recall again the linear system ($\star$):

$$\begin{aligned}\sum_{\tau\in\mathbf{T}} \mathbb{1}\left\{\tau\left(0\right)=d\right\}\left(P\left(T=\tau\right)f_{Y_d|T=\tau}\left(y\right)\right) &= P\left(D=d|Z=0\right)f_{Y|D=d,Z=0}\left(y\right)\\ &\vdots \\ \sum_{\tau\in\mathbf{T}} \mathbb{1}\left\{\tau\left(N_Z\right)=d\right\}\left(P\left(T=\tau\right)f_{Y_d|T=\tau}\left(y\right)\right) &= P\left(D=d|Z=N_Z\right)f_{Y|D=d,Z=N_Z}\left(y\right)\end{aligned} \tag{$\star$}$$

In the notation just defined, we can write ($\star$) as

$$\mathbf{R}_d\left(\boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}\left(y\right)\right)=\mathbf{pf}_{Y|d,Z}\left(y\right) \tag{$\star$}$$

In the applications that motivate this paper, ($\star$) is an underdetermined system, precluding point identification of $\boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}\left(y\right)$.

Informal Outline of Approach

Step 0: Type restriction determines set of possible response-types, $\mathbf{T}$

Step 1: For $d \in \mathcal{D}$, $(\star)$ restricts $\boldsymbol{\pi\rho}_{Y_d|T} := \Big(P(T=\tau) f_{Y_d|T=\tau}(\cdot) : \tau \in \mathbf{T}\Big)$

1a. Imposes restrictions on $\boldsymbol{\pi} := \Big(P(T=\tau) : \tau \in \mathbf{T}\Big)$

1b. Imposes restrictions on the joint values $(P(T=\tau), E(Y_d \mid T=\tau))$

Step 2: Get identified set for $\boldsymbol{\pi}$ by combining restrictions from Step 1a over treatments $d \in \mathcal{D}$

- Incorporate additional assumptions on treatment selection (optional)
- Reject choice restriction iff this set is empty (sharp specification test)

Step 3: Define parameter of interest (e.g., ATE)

- Conduct inference on possible values for this parameter
- Values must be consistent with restrictions from Steps 1 and 2

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Preliminaries: Linear Systems and Convex Polytopes

For any $m \times n$ matrix A, and any m-vector b, define the convex polytope $P_A(b)$ as the set of non-negative solutions to $A\mathbf{x} = \mathbf{b}$

$$P_A(\mathbf{b}) := \left\{\mathbf{x} \in \mathbb{R}^n_{\geq 0} \;\middle|\; A\mathbf{x} = \mathbf{b}\right\} .$$

$P_A(\mathbf{b}) \subseteq \mathbb{R}^n$, but is not n-dimensional (degrees of freedom $< n$)

$\rightarrow$ eliminate some variables to get representation as subset of $\mathbb{R}^{\tilde{n}}$, $\tilde{n} < n$

Define $\tilde{P}_A(\mathbf{b})$ as this representation:

$$\tilde{P}_A(\mathbf{b}) := \left\{\tilde{\mathbf{x}} \in \mathbb{R}^{\tilde{n}} \;\middle|\; \tilde{A}\tilde{\mathbf{x}} \leq \tilde{\mathbf{b}}\right\} ,$$

for some suitable $\tilde{m} \times \tilde{n}$ matrix $\tilde{A}$ and $\tilde{n}$-vector $\tilde{\mathbf{b}}$.

Preliminaries: Support Functions of Convex Sets

Boundary of a convex set can be traced out by its supporting hyperplanes:

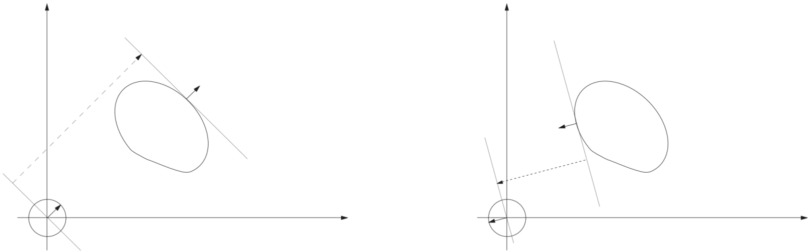

Figure 1: Supporting hyperplanes of a convex set (image: Andres Santos)

For nonempty closed convex set $S \subseteq \mathbb{R}^n$, define the ***support function*** of S, $h_S : \mathbb{R}^n \to \mathbb{R}$, as

$$h_S(\mathbf{c}) := \sup_{\mathbf{x} \in S} \mathbf{c}^\top \mathbf{x}$$

Note also that $-h_S(-\mathbf{c}) = \inf_{\mathbf{x} \in S} \mathbf{c}^\top \mathbf{x}$

Preliminaries: Support Functions of Convex Sets

In the special case where S is a polytope of the form $P_A\left(b\right)$, we will use the following notation:

$$h_A\left(\mathbf{c};\ \mathbf{b}\right) := h_{P_A(\mathbf{b})}(\mathbf{c}) = \sup_{\mathbf{x}\geq 0 : A\mathbf{x}=\mathbf{b}} \mathbf{c}^\top \mathbf{x}$$

We will be especially interested in the polytopes associated with ($\star$),

$$P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}\left(y\right)\right) = \left\{\mathbf{x} \in \mathbb{R}^n_{\geq 0} \,\middle|\, \mathbf{R}_d\mathbf{x} = \mathbf{pf}_{Y|d,Z}\left(y\right)\right\} \ ,$$

and their support functions

$$h_{\mathbf{R}_d}\left(\mathbf{c};\ \mathbf{pf}_{Y|d,Z}\left(y\right)\right) \ .$$

Formal Setup of Model

Our model consists of random variables $(Y, D, Z, \mathbf{V}, \varepsilon_Y)$, defined on a probability space (Ω, Σ, P), and two measurable functions, f_D and f_Y:

$$\text{Treatment Choice Equation:} \quad D = f_D(Z, \mathbf{V}) \;, \tag{1}$$

$$\text{Outcome Equation:} \quad Y = f_Y\left(D, \mathbf{V}, \varepsilon_Y\right) \;, \tag{2}$$

$$\text{Independence Condition:} \quad Z, \mathbf{V}, \varepsilon_Y \text{ are mutually independent} \tag{3}$$

We assume that the observed data, (Y, D, Z), satisfies:

- Y has compact and convex support $\mathcal{Y} \subseteq \mathbb{R}$
- D has finite support $\mathcal{D} := \{0, \ldots, N_D\}$
- Z has finite support $\mathcal{Z} := \{0, \ldots, N_Z\}$

We also need the following technical assumptions:

- $\mathcal{Y}$ is equipped with the Borel σ-field $B\left(\mathcal{Y}\right)$ and atomless measure μ
- For $d \in \mathcal{D}$, $f_Y\left(d, \mathbf{V}, \varepsilon_Y\right)$ is integrable and has a density w.r.t. μ

Formal Setup of Model

Technical Comments

- **Today:** μ can be taken to be the Lebesgue measure
- All equalities and inequalities involving density functions should be understood to hold μ-almost-everywhere
- When we say "all" or "each" $y \in \mathcal{Y}$, we mean "μ-almost-all"
- When we say "there exists a $y \in \mathcal{Y}$" with some property, we mean that the set of $y \in \mathcal{Y}$ with the property has positive μ measure

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Recall: Outline of Approach

Step 0: Type restriction determines set of possible response-types, $\mathbf{T}$

Step 1: For $d \in \mathcal{D}$, ($\star$) restricts $\boldsymbol{\pi\rho}_{Y_d|T}(\cdot) := \left[P(T=\tau)\, f_{Y_d|T=\tau}(\cdot) : \tau \in \mathbf{T}\right]$

1a. Imposes restrictions on $\boldsymbol{\pi} := [P(T=\tau) : \tau \in \mathbf{T}]$

1b. Imposes restrictions on the joint values $(P(T=\tau), E(Y_d \mid T=\tau))$

Step 2: Get identified set for $[P(T=\tau) : \tau \in \mathbf{T}]$ by combining restrictions from Step 1a over treatments $d \in \mathcal{D}$

- Optional: incorporate additional assumptions on treatment selection
- Reject choice restriction iff this set is empty (sharp specification test)

Step 3: Define parameter of interest (e.g. ATE)

- Conduct inference on possible values for this parameter
- Values must be consistent with restrictions from Steps 1 and 2

$\pi\rho_{Y_d|T}(y)$ must be a solution to $(\star)$

Fix $d \in \mathcal{D}$.

For $y \in \mathcal{Y}$, the non-negative solutions to $(\star)$ define the polytope

$$P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) := \left\{\mathbf{x} \in \mathbb{R}^{|\mathbf{T}|}_{\geq 0} \,\middle|\, \mathbf{R}_d\mathbf{x} = \mathbf{pf}_{Y|d,Z}(y)\right\} .$$

Recall that $\pi\rho_{Y_d|T}(y) := \begin{bmatrix} P(T = \tau_0)\, f_{Y_d|T=\tau_0}(y) \\ \vdots \\ P\left(T = \tau_{N_T}\right) f_{Y_d|T=\tau_{N_T}}(y) \end{bmatrix}$ must be a solution to $(\star)$.

$\pi\rho_{Y_d|T}(y)$ must be a solution to $(\star)$

Fix $d \in \mathcal{D}$.

For $y \in \mathcal{Y}$, the non-negative solutions to $(\star)$ define the polytope

$$P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) := \left\{\mathbf{x} \in \mathbb{R}^{|\mathbf{T}|}_{\geq 0} \;\middle|\; \mathbf{R}_d\mathbf{x} = \mathbf{pf}_{Y|d,Z}(y)\right\}.$$

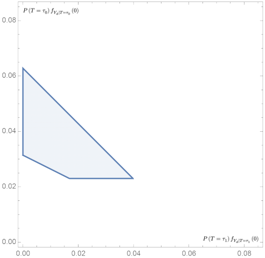
Recall that $\pi\rho_{Y_d|T}(y) := \begin{bmatrix} P(T=\tau_0)\, f_{Y_d|T=\tau_0}(y) \\ \vdots \\ P\left(T=\tau_{N_T}\right) f_{Y_d|T=\tau_{N_T}}(y) \end{bmatrix}$ must be a solution to $(\star)$.

$\rightarrow$ By $(\star)$, we require that $\pi\rho_{Y_d|T}(y) \in P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$ for each $y \in \mathcal{Y}$

$\pi\rho_{Y_d|T}(y)$ must be a solution to ($\star$)

$$P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) := \left\{\mathbf{x} \in \mathbb{R}_{\geq 0}^{|\mathbf{T}|} \;\middle|\; \mathbf{R}_d\mathbf{x} = \mathbf{pf}_{Y|d,Z}(y)\right\} .$$

$\rightarrow$ By ($\star$), we require that $\pi\rho_{Y_d|T}(y) \in P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$ for each $y \in \mathcal{Y}$



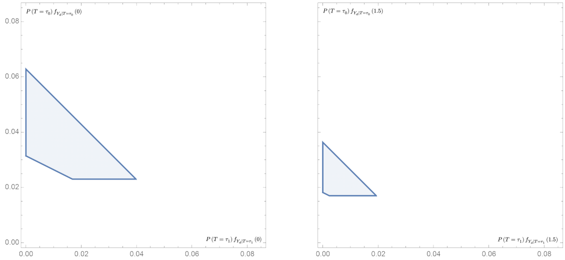
(a) $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(0)\right)$

Figure 2: Hypothetical $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(\cdot)\right)$ in a case with 2 degrees of freedom

$\pi\rho_{Y_d|T}(y)$ must be a solution to $(\star)$

$$P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) := \left\{\mathbf{x} \in \mathbb{R}_{\geq 0}^{|\mathbf{T}|} \;\middle|\; \mathbf{R}_d\mathbf{x} = \mathbf{pf}_{Y|d,Z}(y)\right\} .$$

$\rightarrow$ By $(\star)$, we require that $\pi\rho_{Y_d|T}(y) \in P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$ for each $y \in \mathcal{Y}$



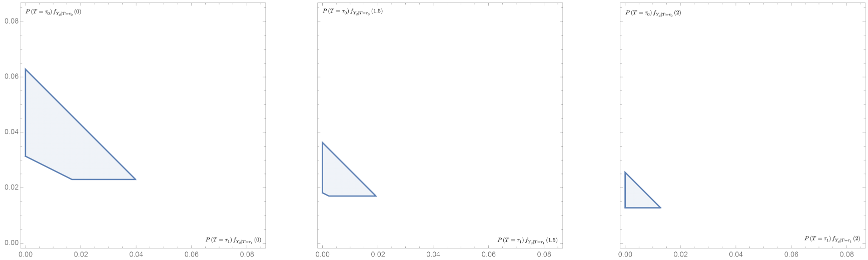
(a) $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(0)\right)$ **(b)** $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(1.5)\right)$

Figure 2: Hypothetical $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(\cdot)\right)$ in a case with 2 degrees of freedom

$\pi\rho_{Y_d|T}(y)$ must be a solution to $(\star)$

$$P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) := \left\{\mathbf{x} \in \mathbb{R}^{|\mathbf{T}|}_{\geq 0} \;\middle|\; \mathbf{R}_d\mathbf{x} = \mathbf{pf}_{Y|d,Z}(y)\right\} .$$

$\rightarrow$ By $(\star)$, we require that $\pi\rho_{Y_d|T}(y) \in P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$ for each $y \in \mathcal{Y}$



(a) $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(0)\right)$ **(b)** $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(1.5)\right)$ **(c)** $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(2)\right)$

Figure 2: Hypothetical $\tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(\cdot)\right)$ in a case with 2 degrees of freedom

$\left(y, \boldsymbol{\pi\rho}_{Y_d|T}(y)\right)$ must lie inside the graph of $y \mapsto P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

Consider the set of all measurable functions $\mathcal{Y} \to \mathbb{R}_{\geq 0}^{|\mathbf{T}|}$ that satisfy ($\star$):

$$\begin{aligned}\Gamma_{\mathbf{T}|d} &:= \left\{\text{measurable } \mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \;\middle|\; \mathbf{g}(y) \in P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right),\ \forall y \in \mathcal{Y}\right\} \\ &= \left\{\text{measurable } \mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \;\middle|\; \mathbf{R}_d\mathbf{g}(y) = \mathbf{pf}_{Y|d,Z}(y),\ \forall y \in \mathcal{Y}\right\}\end{aligned}$$

$\to$ we must have $\boldsymbol{\pi\rho}_{Y_d|T} \in \Gamma_{\mathbf{T}|d}$.

$\left(y, \pi\rho_{Y_d|T}(y)\right)$ must lie inside the graph of $y \mapsto P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

$$\Gamma_{\mathbf{T}|d} := \left\{\text{measurable } \mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \;\middle|\; \mathbf{R}_d\mathbf{g}(y) = \mathbf{pf}_{Y|d,Z}(y)\,,\ \forall y \in \mathcal{Y}\right\}$$

$\rightarrow$ we must have $\pi\rho_{Y_d|T} \in \Gamma_{\mathbf{T}|d}$.

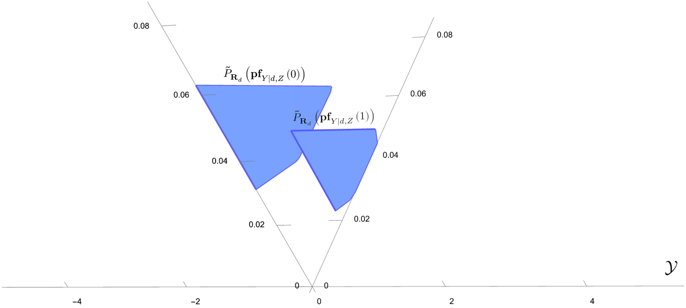


Figure 3: Hypothetical graph of the correspondence $y \mapsto \tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

$\left(y, \pi\rho_{Y_d|T}(y)\right)$ must lie inside the graph of $y \mapsto P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

$$\Gamma_{\mathbf{T}|d} := \left\{\text{measurable } \mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \;\middle|\; \mathbf{R}_d\mathbf{g}(y) = \mathbf{pf}_{Y|d,Z}(y)\,,\ \forall y \in \mathcal{Y}\right\}$$

$\rightarrow$ we must have $\pi\rho_{Y_d|T} \in \Gamma_{\mathbf{T}|d}$.

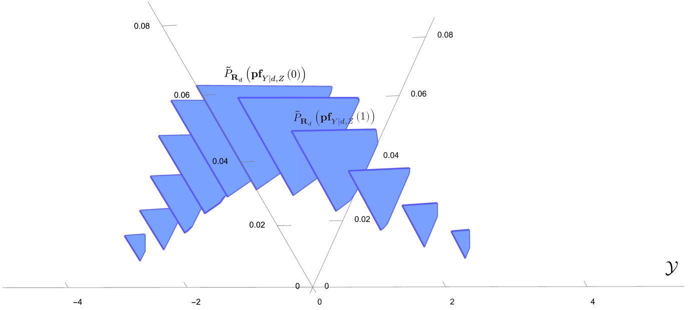


Figure 3: Hypothetical graph of the correspondence $y \mapsto \tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

$\left(y, \pi\rho_{Y_d|T}(y)\right)$ must lie inside the graph of $y \mapsto P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

$$\Gamma_{\mathbf{T}|d} := \left\{\text{measurable } \mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \;\middle|\; \mathbf{R}_d\mathbf{g}(y) = \mathbf{pf}_{Y|d,Z}(y)\,,\ \forall y \in \mathcal{Y}\right\}$$

$\rightarrow$ we must have $\pi\rho_{Y_d|T} \in \Gamma_{\mathbf{T}|d}$.

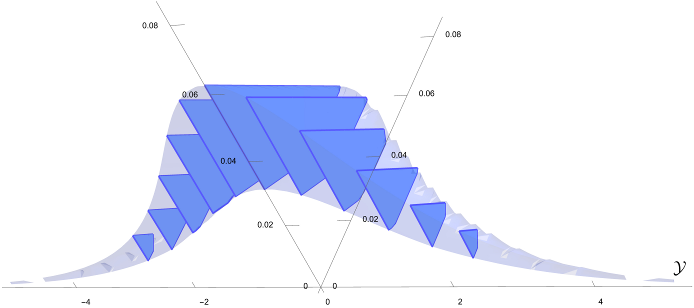


Figure 3: Hypothetical graph of the correspondence $y \mapsto \tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

$\left(y, \pi\rho_{Y_d|T}(y)\right)$ must lie inside the graph of $y \mapsto P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

$$\Gamma_{\mathbf{T}|d} := \left\{\text{measurable } \mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \;\middle|\; \mathbf{R}_d\mathbf{g}(y) = \mathbf{pf}_{Y|d,Z}(y)\,,\ \forall y \in \mathcal{Y}\right\}$$

$\to$ we must have $\pi\rho_{Y_d|T} \in \Gamma_{\mathbf{T}|d}$.

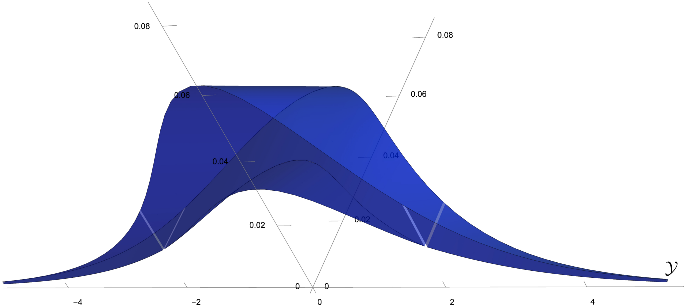


Figure 3: Hypothetical graph of the correspondence $y \mapsto \tilde{P}_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$

Functions in $\Gamma_{\mathbf{T}|d}$ are "candidates" for $\pi\rho_{Y_d|T}$

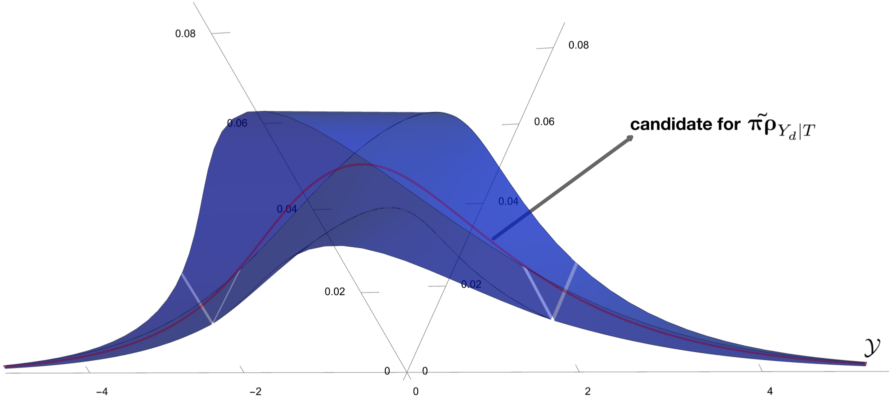


Figure 4: Hypothetical candidate $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

Functions in $\Gamma_{\mathbf{T}|d}$ are "candidates" for $\pi\rho_{Y_d|T}$

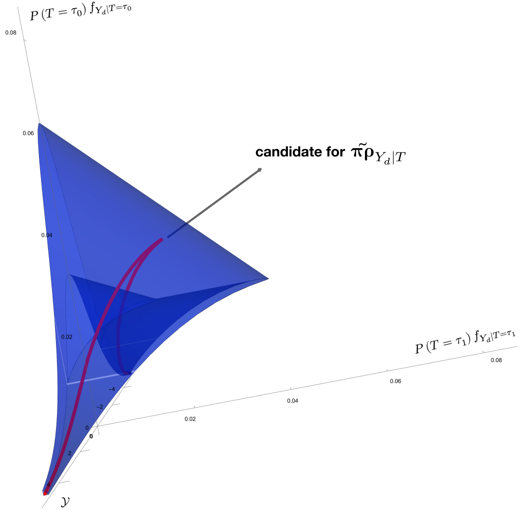
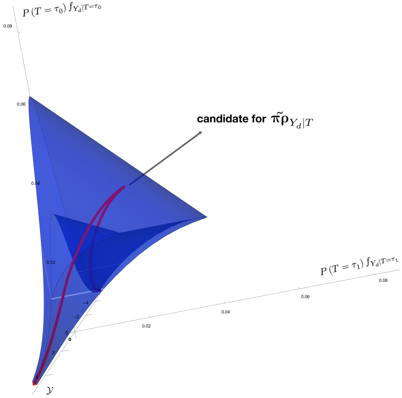


Figure 5: Hypothetical candidate $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

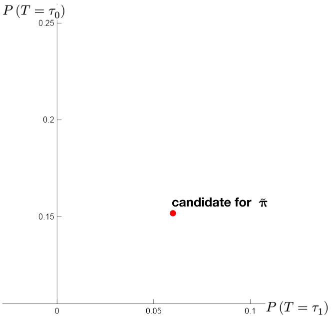
Note that if $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$ is a candidate for $\boldsymbol{\pi\rho}_{Y_d|T}$, then $\int_{\mathcal{Y}} \mathbf{g}(y) \; d\mu(y)$ is a candidate for $\boldsymbol{\pi}$, the vector of response-type probabilities. Precisely,

$$\mathbf{g} = \boldsymbol{\pi\rho}_{Y_d|T} \implies \int_{\mathcal{Y}} \mathbf{g}(y) \; d\mu(y) = \boldsymbol{\pi}$$

If $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$, then $\int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu(y)$ is a candidate for π



(a) $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

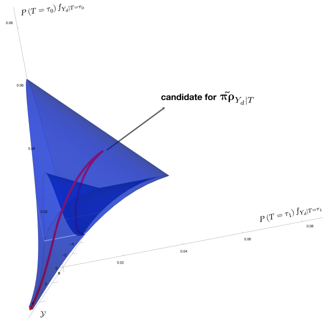


(b) $\int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu(y)$

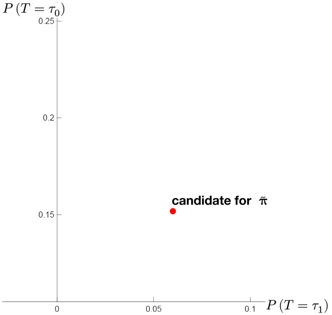
Candidates for π

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for π:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu\,(y) \,\middle|\, \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$



(a) $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

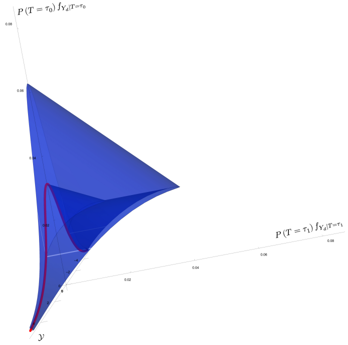


(b) $\int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu(y)$

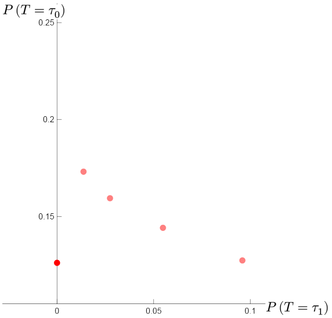
Candidates for π

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for π:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu\left(y\right) \,\middle|\, \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$



(a) $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

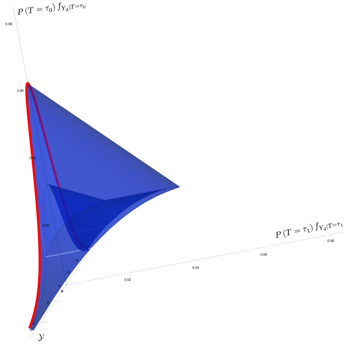


(b) $\int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu(y)$

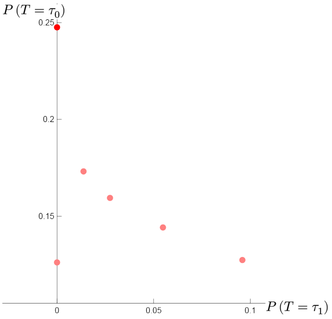
Candidates for π

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for π:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu\,(y) \,\middle|\, \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$



(a) $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

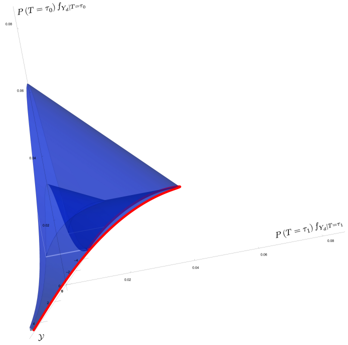


(b) $\int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu(y)$

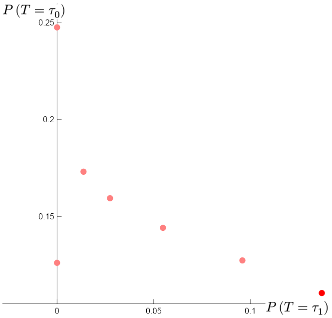
Candidates for π

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for π:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu\,(y) \,\middle|\, \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$



(a) $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

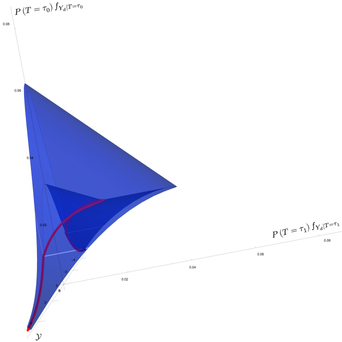


(b) $\int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu(y)$

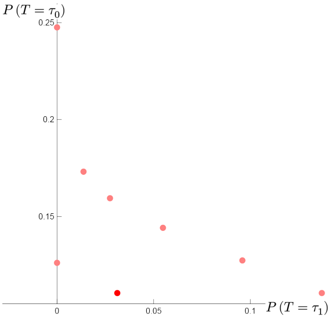
Candidates for π

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for π:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu\,(y) \,\middle|\, \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$



(a) $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

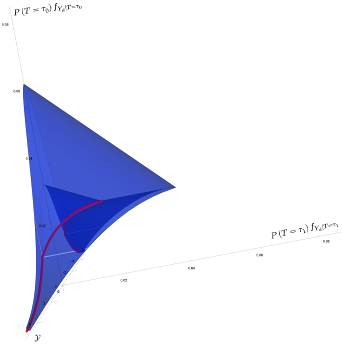


(b) $\int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu(y)$

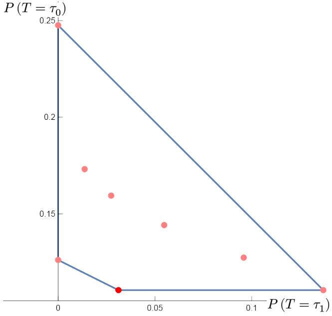
Candidates for π

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for π:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu\,(y) \;\middle|\; \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$



(a) $\mathbf{g} \in \Gamma_{\mathbf{T}|d}$

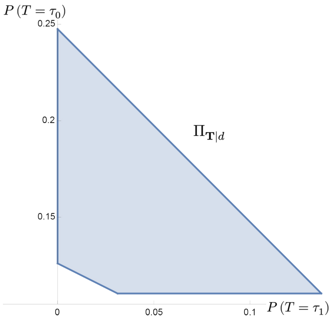


(b) $\int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu(y)$

Candidates for π

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for π:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\; d\mu\,(y) \;\middle|\; \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$



Candidates for $\boldsymbol{\pi}$

Define $\Pi_{\mathbf{T}|d}$ as the set of all such candidates for $\boldsymbol{\pi}$:

$$\Pi_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\ d\mu\,(y) \;\middle|\; \mathbf{g} \in \Gamma_{\mathbf{T}|d} \right\} .$$

$\rightarrow \Pi_{\mathbf{T}|d}$ is a convex set, and its boundary can be obtained by integrating along the boundary of $P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}\,(y)\right)$

Comments

- $\Pi_{\mathbf{T}|d} = \emptyset$ iff there is $y \in \mathcal{Y}$ such that $P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}\,(y)\right) = \emptyset$
- We must have $\boldsymbol{\pi} \in \Pi_{\mathbf{T}|d}$ for every $d \in \mathcal{D}$
- $\Pi_{\mathbf{T}|d}$ is the *Minkowski integral* of the map $y \mapsto P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}\,(y)\right)$
  - Continuous analogue to Minkowski sum of polytopes
  - Close connection to *Aumann expectation* from theory of random sets (Aumann, 1965; Artstein and Vitale, 1975)

Candidates for π

Lemma 1

Suppose $\mathbf{T}$ is a type restriction such that $P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) \neq \emptyset$ for all $y \in \mathcal{Y}$. Then, for all $\mathbf{c} \in \mathbb{R}^{|\mathbf{T}|}$

$$h_{\Pi_{\mathbf{T}|d}}(\mathbf{c}) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(\mathbf{c};\ \mathbf{pf}_{Y|d,Z}(y)\right)\ d\mu(y)\ .$$

Further, $\Pi_{\mathbf{T}|d}$ is a convex polytope.

Proof sketch: We can view the correspondence $y \mapsto P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)$ as a *polytope bundle* and $\Pi_{\mathbf{T}|d}$ as its *Minkowski integral*; the result now follows from general convex geometry results about polytope bundles (see Proposition 1.2 in Billera and Sturmfels (1992)).

Recall: Outline of Approach

Step 0: Type restriction determines set of possible response-types, $\mathbf{T}$

Step 1: For $d \in \mathcal{D}$, $(\star)$ restricts $\pi\rho_{Y_d|T}(\cdot) := \left[P(T=\tau) f_{Y_d|T=\tau}(\cdot) : \tau \in \mathbf{T}\right]$

1a. Imposes restrictions on $\pi := [P(T=\tau) : \tau \in \mathbf{T}]$

1b. Imposes restrictions on the joint values $(P(T=\tau), E(Y_d \mid T=\tau))$

Step 2: Get identified set for $\pi := \left(P(T=\tau) : \tau \in \mathbf{T}\right)$ by combining restrictions from Step 1a over treatments $d \in \mathcal{D}$

- Optional: incorporate additional assumptions on treatment selection
- Reject choice restriction iff this set is empty (sharp specification test)

Step 3: Define parameter of interest (e.g. ATE)

- Conduct inference on possible values for this parameter
- Values must be consistent with restrictions from Steps 1 and 2

Identified Set for Response-Type Probabilities

Let $\mathcal{Q}_{\mathbf{T}}$ be the set of joint distributions $\left(Y_0, \ldots, Y_{N_D}, D_0, \ldots, D_{N_Z}, Z\right)$ that are consistent with the model, type restriction $\mathbf{T}$, and the observed data.

Define the identified set for $\boldsymbol{\pi} = \Big(P\left(T = \tau\right) : \tau \in \mathbf{T}\Big)$, denoted $\Pi_{\mathbf{T}}$, as

$$\Pi_{\mathbf{T}} := \left\{\Big(Q\left(T = \tau\right) : \tau \in \mathbf{T}\Big) : Q \in \mathcal{Q}_{\mathbf{T}}\right\}.$$

Theorem 1 (Identified Set for Response-Type Probabilities)

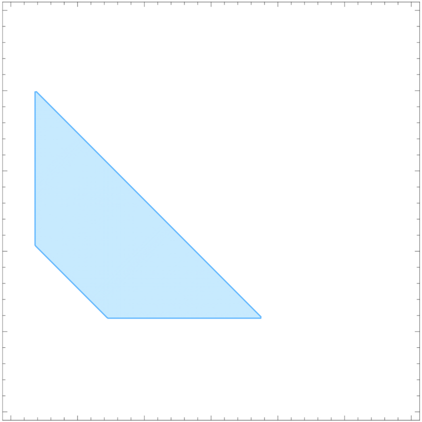
For any type restriction $\mathbf{T}$, $\Pi_{\mathbf{T}}$ is a compact convex polytope and

$$\Pi_{\mathbf{T}} = \bigcap_{d \in \mathcal{D}} \Pi_{\mathbf{T}|d}\,.$$

In particular, there exists matrix $\bar{\mathbf{R}}$ and vector $\bar{\mathbf{b}}$ such that

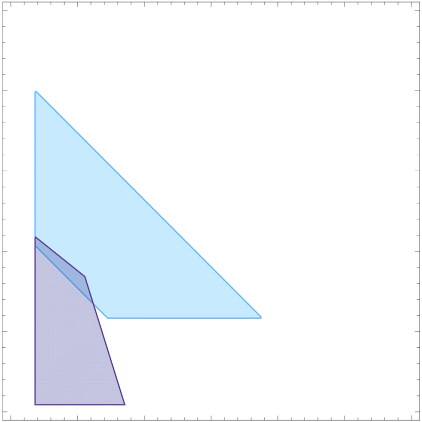
$$\Pi_{\mathbf{T}} = \left\{\mathbf{x} \in \mathbb{R}_{\geq 0}^{|\mathbf{T}|} \;\middle|\; \bar{\mathbf{R}}\mathbf{x} = \bar{\mathbf{b}}\right\}.$$

(Stylized) Example with 3 Treatments



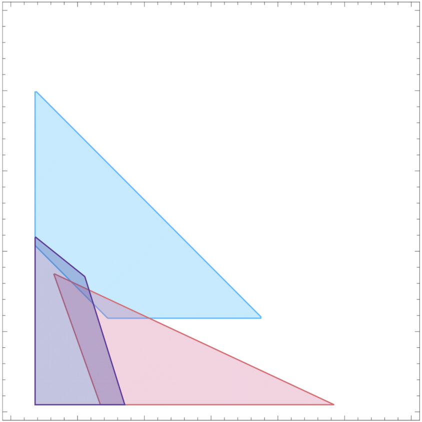
(a) Hypothetical $\Pi_{\mathbf{T}|1}$

(Stylized) Example with 3 Treatments



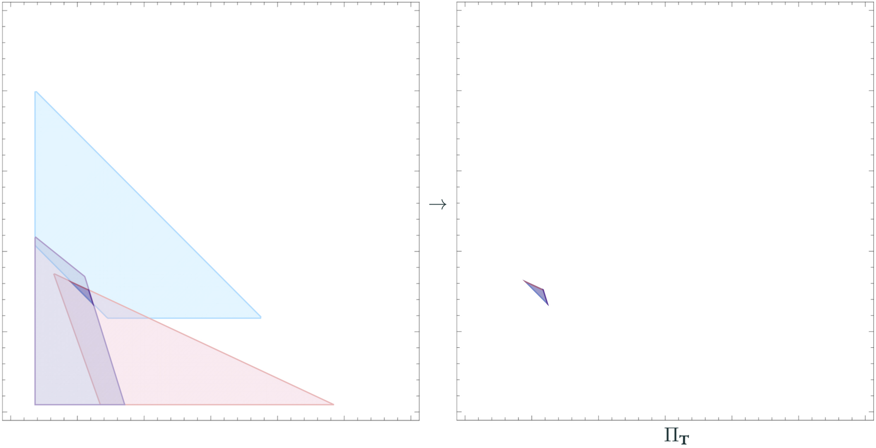
(a) Hypothetical $\Pi_{\mathbf{T}|1}, \Pi_{\mathbf{T}|2}$

(Stylized) Example with 3 Treatments



(a) Hypothetical $\Pi_{\mathbf{T}|1}, \Pi_{\mathbf{T}|2}, \Pi_{\mathbf{T}|3}$

(Stylized) Example with 3 Treatments



(a) Hypothetical $\Pi_{\mathbf{T}} := \bigcap_{d \in \mathcal{D}} \Pi_{\mathbf{T}|d}$

▸ Basic Example

Implications of Theorem 1: Specification Test

Corollary 1 (Sharp Misspecification Test)

For any type restriction $\mathbf{T}$, $\mathcal{Q}_{\mathbf{T}} \neq \emptyset$ if and only if $\Pi_{\mathbf{T}} \neq \emptyset$.

Comments

- Corollary 1 provides joint testable implications for type restrictions and random assignment of Z
  - Indeed, setting $\mathbf{T} = \mathcal{D}^{|\mathcal{Z}|}$ in Corollary 1 produces a sharp test for random assignment of Z
- More generally, a given choice restriction is rejected by the data iff there is no point in $\Pi_{\mathbf{T}}$ that satisfies the choice restriction, where $\mathbf{T}$ is the set of types allowed by the choice restriction

Implications of Theorem 1: Specification Test

...More Comments

- Corollary 1 requires (i) $\Pi_{\mathbf{T}|d} \neq \emptyset$ for each $d \in \mathcal{D}$, and (ii) $\bigcap_{d\in\mathcal{D}} \Pi_{\mathbf{T}|d} \neq \emptyset$
- $\Pi_{\mathbf{T}|d} \neq \emptyset$ iff $\mathbf{pf}_{Y|d,Z}(y) \in \text{cone}(\mathbf{R}_d)$ for all $y \in \mathcal{Y}$
  - $\text{cone}(\mathbf{R}_d)$ is the polyhedral cone generated by positive linear combinations of the columns of $\mathbf{R}_d$
  - For each d, $\Pi_{\mathbf{T}|d} \neq \emptyset$ can be tested using Fang and Seo (2021)
  - (i) is automatically satisfied in some cases; e.g., $\mathbf{T} = \mathcal{D}^{|\mathcal{Z}|}$
- Given condition (i), we can test condition (ii) using Fang, Santos, Shaikh, and Torgovitsky (2023), with an estimator based on Lemma 1
- We also propose an approach that uses the halfspace representation of $\text{cone}(\mathbf{R}_d)$ (Minkowski-Weyl theorem) to obtain a joint test for (i) and (ii) based on conditional moment inequalities ▸ More Details

Implications of Theorem 1: Potential Outcome Densities

Corollary 2 (Identified Set for Potential Outcome Densities)

For any type restriction $\mathbf{T}$ and treatment $d \in \mathcal{D}$, the identified set for $\mathbf{f}_{Y_d|T} := \left[f_{Y_d|T=\tau} : \tau \in \mathbf{T} \right]$, the vector of Y_d-densities conditional on type, is given by

$$\bigcup_{\boldsymbol{\pi} \in \Pi_{\mathbf{T}}} \left\{ \mathbf{f} : \mathcal{Y} \to \mathbb{R}_{\geq 0}^{|\mathbf{T}|} \;\middle|\; \int_{\mathcal{Y}} \mathbf{f}(y)\, d\mu\left(y\right) = \mathbf{1}, \boldsymbol{\pi} \odot \mathbf{f} \in \Gamma_{\mathbf{T}|d} \right\}$$

Note: Here $\odot$ denotes the element-wise product of two vectors (Hadamard product), i.e. $\boldsymbol{\pi} \odot \mathbf{f} := \mathrm{diag}\left(\boldsymbol{\pi}\right) \mathbf{f}$

Recall: Outline of Approach

Step 0: Type restriction determines set of possible response-types, $\mathbf{T}$

Step 1: For $d \in \mathcal{D}$, ($\star$) restricts $\boldsymbol{\pi\rho}_{Y_d|T}(\cdot) := \left[P(T=\tau)\, f_{Y_d|T=\tau}(\cdot) : \tau \in \mathbf{T}\right]$

1a. Imposes restrictions on $\boldsymbol{\pi} := [P(T=\tau) : \tau \in \mathbf{T}]$

1b. Imposes restrictions on the joint values
$(P(T=\tau), P(T=\tau)\, E(Y_d \mid T=\tau))$

Step 2: Get identified set for $[P(T=\tau) : \tau \in \mathbf{T}]$ by combining restrictions from Step 1a over treatments $d \in \mathcal{D}$

- Optional: incorporate additional assumptions on treatment selection
- Reject choice restriction iff this set is empty (sharp specification test)

Step 3: Define parameter of interest (e.g. ATE)

- Conduct inference on possible values for this parameter
- Values must be consistent with restrictions from Steps 1 and 2

Counterfactual Means

Fix $d \in \mathcal{D}$. For any $y \in \mathcal{Y}$, consider the following polytope

$$\begin{aligned}\mathcal{E}_d(y) &:= \left\{ (\mathbf{x}_1, \mathbf{x}_2) \in \mathbb{R}^{|\mathbf{T}|} \times \mathbb{R}^{|\mathbf{T}|}_{\geq 0} \;\middle|\; \mathbf{x}_1 = y\mathbf{x}_2,\ \mathbf{R}_d\mathbf{x}_2 = \mathbf{pf}_{Y|d,Z}(y) \right\} \\ &= \left\{ (\mathbf{x}_1, \mathbf{x}_2) \in \mathbb{R}^{|\mathbf{T}|} \times \mathbb{R}^{|\mathbf{T}|}_{\geq 0} \;\middle|\; \mathbf{x}_1 = y\mathbf{x}_2,\ \mathbf{x}_2 \in P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) \right\}.\end{aligned}$$

We can view any $\mathbf{x} \in \mathcal{E}_d(y)$ as a candidate for the stacked vector

$$\left(y\boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}(y)^\top, \boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}(y)^\top \right)$$

Also, define $\boldsymbol{\pi}\boldsymbol{\theta}_d$ as the vector of weighted conditional means

$$\boldsymbol{\pi}\boldsymbol{\theta}_d := \left(P(T = \tau) E(Y_d \mid T = \tau) : \tau \in \mathbf{T} \right)$$

Candidates for the function $y \mapsto \left(y\boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}(y)^\top, \boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}(y)^\top\right)$

Consider the set of all measurable functions $\mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \times \mathbb{R}^{|\mathbf{T}|}_{\geq 0}$ such that $\mathbf{g}(y) \in \mathcal{E}_d(y)$ for every $y \in \mathcal{Y}$:

$$\Gamma^{\mathcal{E}}_{\mathbf{T}|d} := \left\{\text{measurable } \mathbf{g} : \mathcal{Y} \to \mathbb{R}^{|\mathbf{T}|} \times \mathbb{R}^{|\mathbf{T}|} \mid \mathbf{g}(y) \in \mathcal{E}_d(y),\ \forall y \in \mathcal{Y}\right\}$$

Now, note that

$$\mathbf{g}(y) = \left(y\boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}(y)^\top, \boldsymbol{\pi}\boldsymbol{\rho}_{Y_d|T}(y)^\top\right)\ \forall y \in \mathcal{Y}$$

$$\implies \int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu(y) = \left(\boldsymbol{\pi}\boldsymbol{\theta}_d^\top, \boldsymbol{\pi}^\top\right) = \begin{bmatrix} P(T=\tau_0)\, E(Y_d \mid T=\tau_0) \\ \vdots \\ P\left(T=\tau_{N_T}\right) E\left(Y_d \mid T=\tau_{N_T}\right) \\ P(T=\tau_0) \\ \vdots \\ P\left(T=\tau_{N_T}\right) \end{bmatrix}$$

Candidates for $(\boldsymbol{\theta}_d^\top, \boldsymbol{\pi}^\top)$

Define $\Theta_{\mathbf{T}|d}$ as the set of all such candidates for $(\boldsymbol{\pi}\boldsymbol{\theta}_d^\top, \boldsymbol{\pi}^\top)$:

$$\Theta_{\mathbf{T}|d} := \left\{ \int_{\mathcal{Y}} \mathbf{g}(y)\, d\mu(y) \;\middle|\; \mathbf{g} \in \Gamma^{\mathcal{E}}_{\mathbf{T}|d} \right\} .$$

$\Theta_{\mathbf{T}|d}$ is not generally a polytope, but it is always convex.

Lemma 2

Suppose $\mathbf{T}$ is a type restriction such that $P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right) \neq \emptyset$ for all $y \in \mathcal{Y}$. Then, $\Theta_{\mathbf{T}|d}$ is convex, and for all $(\mathbf{c}_1, \mathbf{c}_2) \in \mathbb{R}^{|\mathbf{T}|} \times \mathbb{R}^{|\mathbf{T}|}$,

$$h_{\Theta_{\mathbf{T}|d}}(\mathbf{c}_1, \mathbf{c}_2) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(y\mathbf{c}_1 + \mathbf{c}_2;\; \mathbf{pf}_{Y|d,Z}(y)\right)\, d\mu(y) .$$

1. Informal Overview of Approach
2. Preliminaries and Formal Setup
3. Main Identification Results
4. Basic Outline of Inference Strategy
5. Conclusion

Recall: Outline of Approach

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1a. Imposes restrictions on $\pi := [P(T=\tau) : \tau \in \mathbf{T}]$

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Parameter of Interest

Let some arbitrary weight vector $\mathbf{c} \in \mathbb{R}^{|\mathbf{T}|}$ be given, and suppose the target parameter β is defined as the following weighted average

$$\beta := \frac{\mathbf{c}^\top \Big(P\left(T = \tau\right) E\left(Y_d \mid T = \tau\right) : \tau \in \mathbf{T} \Big)}{\mathbf{c}^\top \Big(P\left(T = \tau\right) : \tau \in \mathbf{T} \Big)} .$$

For a given $\gamma \in \mathbb{R}$, we now consider the problem of testing if γ is in the identified set for β.

Comments

- This includes testing $E\left(Y_d \mid T \subseteq \mathbf{T}_\beta\right) = \gamma$ (for some subset $\mathbf{T}_\beta \subseteq \mathbf{T}$), as a special case
- The approach for inference on treatment effects is very similar, focusing on this case only for simplicity

Inclusion Test

Denote by $\Theta_{\mathbf{T}}(\mathbf{c})$ the identified set for β. Corollary 2 implies that

$$\Theta_{\mathbf{T}}(\mathbf{c}) = \left\{ \frac{\mathbf{c}^\top \boldsymbol{\pi} \boldsymbol{\theta}_d}{\mathbf{c}^\top \boldsymbol{\pi}} : (\boldsymbol{\pi}\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\mathbf{T}|d},\ \boldsymbol{\pi} \in \Pi_{\mathbf{T}} \right\}$$
$$= \left\{ b \in \mathbb{R} : \exists\ (\boldsymbol{\pi}\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\mathbf{T}|d} \text{ s.t. } \boldsymbol{\pi} \in \Pi_{\mathbf{T}}, \mathbf{c}^\top \boldsymbol{\pi}\boldsymbol{\theta}_d = b\mathbf{c}^\top \boldsymbol{\pi} \right\}.$$

We are interested in the following hypothesis testing problem:

$$H_0 : \gamma \in \Theta_{\mathbf{T}}(\mathbf{c}) \text{ against } H_A : \gamma \notin \Theta_{\mathbf{T}}(\mathbf{c}) \tag{4}$$

Define the following closed convex polytope

$$\Lambda_{\mathbf{T}}(\mathbf{c};\gamma) := \left\{ (\mathbf{u}, \boldsymbol{\pi}) \in \mathbb{R}^{|\mathbf{T}|} \times \mathbb{R}^{|\mathbf{T}|} : \boldsymbol{\pi} \in \Pi_{\mathbf{T}},\ \mathbf{c}^\top \mathbf{u} = \gamma \mathbf{c}^\top \boldsymbol{\pi} \right\},$$

and note that $\gamma \in \Theta_{\mathbf{T}}(\mathbf{c}) \iff \Lambda_{\mathbf{T}}(\mathbf{c};\gamma) \cap \Theta_{\mathbf{T}|d} \neq \emptyset$.

Conclusion

Summary

- General method for inference on mean parameters in IV models with many treatments and/or instruments
- Can incorporate a broad class of treatment choice models
- New characterization of identified set for potential outcome distributions in such models
- Allows for sharp specification testing

In Progress

- Formal inference and estimation results
- Monte Carlo experiments
- Developing a Stata/R/Python package
- Additional guidance on when bounds are likely to be informative

THANKS :)

Additional Parameters: Conditioning on Outcomes

The results in this paper also provide a new characterization of the identified set for the *joint* distribution of potential outcomes, allowing us to consider parameters of the form

$$\beta = E\left(Y_d - Y_{d'} \mid Y_{d''} \in \mathcal{Y}_\beta\right) ,$$

for some subset $\mathcal{Y}_\beta \subseteq \mathcal{Y}$.

Example: In education settings, we may be interested in evaluating the impact of a new curriculum on those who would underperform under the old curriculum; in this case, we may use $\beta = E\left(Y_1 - Y_0 \mid Y_0 < c\right)$

Will not focus on this parameter today.

Back

Example (Theorem 1)

Basic Example ($Z \in \{0,1\}$, $D \in \{0,1\}$, no choice restrictions) :

For $d = 1$, ($\star$) implies

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \boldsymbol{\pi}\boldsymbol{\rho}_{Y_1|T}(y) = \begin{bmatrix} P(D=1|Z=0)\, f_{Y|D=1,Z=0}(y) \\ P(D=1|Z=1)\, f_{Y|D=1,Z=1}(y) \end{bmatrix}$$

$$\implies \begin{bmatrix} 1 \\ 1 \end{bmatrix} P(T=(1,1))\, f_{Y_1|T=(1,1)}(y) \leq \begin{bmatrix} P(D=1|Z=0)\, f_{Y|D=1,Z=0}(y) \\ P(D=1|Z=1)\, f_{Y|D=1,Z=1}(y) \end{bmatrix}$$

The support function for the 1-dimensional polytope (i.e. interval) defined by the non-negative solutions to the above system is simply the map

$$c \mapsto \begin{cases} c\left(\min_{z\in\{0,1\}} \left\{P(D=1|Z=z)\, f_{Y|D=1,Z=z}(y)\right\}\right) & \text{if } c \geq 0 \\ 0 & \text{if } c < 0 \end{cases}$$

Example (Theorem 1)

$\Pi_{\mathbf{T}|1}$ consists of all $\pi \geq 0$ such that

$$\pi_{(1)} \leq \int_{\mathcal{Y}} \min_{z\in\{0,1\}} \left\{ P\left(D=1|Z=z\right) f_{Y|D=1,Z=z}\left(y\right) \right\} \, d\mu\left(y\right) ,$$

$$\pi_{(1)} + \pi_{(3)} = P\left(D=1|Z=0\right) = \int_{\mathcal{Y}} P\left(D=1|Z=0\right) f_{Y|D=1,Z=0}\left(y\right) \, d\mu\left(y\right) ,$$

$$\pi_{(1)} + \pi_{(2)} = P\left(D=1|Z=1\right) = \int_{\mathcal{Y}} P\left(D=1|Z=1\right) f_{Y|D=1,Z=1}\left(y\right) \, d\mu\left(y\right) .$$

Similarly, $\Pi_{\mathbf{T}|0}$ consists of all $\pi \geq 0$ such that

$$\pi_{(4)} \leq \int_{\mathcal{Y}} \min_{z\in\{0,1\}} \left\{ P\left(D=0|Z=z\right) f_{Y|D=0,Z=z}\left(y\right) \right\} \, d\mu\left(y\right) ,$$

$$\pi_{(2)} + \pi_{(4)} = P\left(D=0|Z=0\right) ,$$

$$\pi_{(3)} + \pi_{(4)} = P\left(D=0|Z=1\right) .$$

$\Pi_{\mathbf{T}}$ consists of all $\pi \geq 0$ that satisfy the 6 conditions above.

Back to Theorem 1

A Preliminary: Basic Solutions of Linear Programs

Consider the linear program (LP):

$$\max\ \mathbf{c}^\top \mathbf{x} \qquad \text{subject to} \quad A\mathbf{x} = \mathbf{b},\ \mathbf{x} \geq \mathbf{0}$$

For each subset $B \subseteq \{1, \dots, n\}$ of columns of A, let A_B be the sub-matrix formed by the columns in B. If A_B^{-1} is invertible, we will say that B is a *basis*.

For any basis $B = \{j_1, \dots, j_m\}$, with $j_1 < j_2 < \cdots < j_m$, such that $A_B^{-1}\mathbf{b} \geq 0$, we can construct a feasible point $\mathbf{x}$, as follows:
set $\mathbf{x}_i = 0$ for all $i \notin B$, and set each $\mathbf{x}_{j_k}$ to the k-the value of $A_B^{-1}\mathbf{b}$.

In an abuse of notation, we will simply use $A_B^{-1}\mathbf{b}$ to denote the feasible point (vertex) constructed this way.

Suppose the LP has at least one solution, and that $\mathbf{c}'\mathbf{x}$ is bounded over the feasible space. Then, there always exists a basis $B^*(\mathbf{c}; \mathbf{b})$ such that an **optimal** solution to the above LP is given by $A_{B^*(\mathbf{c};\mathbf{b})}^{-1}\mathbf{b}$.

Back to Corollary 1 Back to Conclusion

Computation of Support Functions (Informal Outline)

The tractability of the test proposed by Corollary 1 (and of all other results in this paper) depends on being able to estimate, for given $\mathbf{c} \in \mathbb{R}^{|\mathbf{T}|}$,

$$\int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(\mathbf{c};\ \mathbf{pf}_{Y|d,Z}(y)\right)\ d\mu(y)\ . \tag{6}$$

Formal estimation results are pending.

- When there are only a "few" instrument-levels, $h_{\mathbf{R}_d}(\mathbf{c};\ \cdot)$ can be solved symbolically (either by hand or using, e.g., Mathematica)
  - Given a symbolic solution, a consistent estimator (with non-degenerate asymptotic distribution) for (6) can be obtained by plugging in a suitable kernel density estimator for $\mathbf{pf}_{Y|d,Z}(y)$

Computation of Support Functions (Informal Outline)

When there are "many" instrument-levels, the estimate $h_{\mathbf{R}_d}\left(\mathbf{c};\ \widehat{\mathbf{pf}}_{Y|d,Z}(y)\right)$ can be approximated.

We propose the following approach (informal outline):

1. Fix a grid $\inf \mathcal{Y} =: y_0 \leq \cdots \leq y_N := \sup \mathcal{Y}$.
2. At each grid point, find optimal *basis*, $B^*(y_n) := B^*\left(\mathbf{c}; \widehat{\mathbf{pf}}_{Y|d,Z}(y_n)\right)$, to get the solution $\mathbf{R}^{-1}_{d,B^*(y_n)}\widehat{\mathbf{pf}}_{Y|d,Z}(y_n)$.
3. For $y_n \leq y \leq y_{n+1}$, estimate $h_{\mathbf{R}_d}\left(\mathbf{c};\ \widehat{\mathbf{pf}}_{Y|d,Z}(y)\right)$ as

$$\max\left\{\mathbf{c}^\top \mathbf{R}^{-1}_{d,B^*(y_n)}\widehat{\mathbf{pf}}_{Y|d,Z}(y),\mathbf{c}^\top \mathbf{R}^{-1}_{d,B^*(y_{n+1})}\widehat{\mathbf{pf}}_{Y|d,Z}(y)\right\}$$

Computation of Support Functions (Informal Outline)

- Generally, tests based on the above approach will be invalid
  - Final step is an approximation based on the *primal* solution; this leads to an "inner" approximation of the polytope (too optimistic)
- We can instead approximate (6) using the ***dual*** LP solution, which will lead to conservative inference
  - Dual approximation is an "outer" approximation of the polytope (by weak duality theorem)
- The difference between the two estimates provides a bound on the approximation error at each point (more grid points can be added around points with large error until this error is sufficiently small)
- Always possible to obtain an exact estimate in finitely many steps
- Computation of $\int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(y\mathbf{v}_1 + \mathbf{v}_2;\ \mathbf{pf}_{Y|d,Z}(y)\right)\ d\mu(y)$ can proceed using a similar approach, by replacing the LP here with a suitable parametric LP (parameterized by y)

Specification Test Implementation (Informal Outline)

Implementing the test suggested by Corollary 1 requires us to test (i) $\Pi_{\mathbf{T}|d} \neq \emptyset$ for each $d \in \mathcal{D}$, and (ii) $\bigcap_{d \in \mathcal{D}} \Pi_{\mathbf{T}|d} \neq \emptyset$

To test (i), we need to test that $\mathbf{pf}_{Y|d,Z}(y) \in \operatorname{cone}(\mathbf{R}_d)$ for all $y \in \mathcal{Y}$.

Let $\mathbf{A}_d$ be a matrix that represents a halfspace representation of $\operatorname{cone}(\mathbf{R}_d)$, so that

$$\operatorname{cone}(\mathbf{R}_d) := \{\mathbf{b} : \exists\, \mathbf{x} \geq 0,\ \mathbf{R}_d\mathbf{x} = \mathbf{b}\} = \{\mathbf{b} : \mathbf{A}_d\mathbf{b} \geq 0\}$$

Now, testing (i) is equivalent to testing $\inf_{y \in \mathcal{Y}} \mathbf{A}_d \mathbf{pf}_{Y|d,Z}(y) \geq 0$.

Specification Test Implementation (Informal Outline)

$\mathbf{A}_d \mathbf{pf}_{Y|d,Z}(y) \geq 0$ represents a set of linear inequalities.

For each row $z \in \{0, \dots, N_Z\}$, the corresponding linear inequality can be represented as a conditional moment inequality of the form

$$E\left[\sum_{k=1}^{N_Z} \tilde{c}_k(z) \mathbb{1}\{D = d, Z = k\} \,\middle|\, Y = y\right] \geq 0,$$

using arguments analogous to those used in Mourifié and Wan (2017)[2].

Define $\gamma_d^j(y) := E\left[\sum_{k=1}^{N_Z} \tilde{c}_k(j) \mathbb{1}\{D = d, Z = k\} \,\middle|\, Y = y\right]$, and define

$$\gamma := \inf_{(y,d,z) \in \mathcal{Y} \times \mathcal{D} \times \mathcal{Z}} \gamma_d^z(y) \quad .$$

[2]Here, the $\tilde{c}_k(z)$ are known functions of P_Z that can be easily pre-estimated, and then treated as constants in the remainder of the procedure (see Appendix A.1 in Mourifié and Wan (2017))

Specification Test Implementation (Informal Outline)

To test (ii), let $\bar{\mathbf{R}}$ and $\bar{\mathbf{b}}$ be as given in Theorem 1, so that

$$\Pi_{\mathbf{T}} = \left\{ \mathbf{x} \in \mathbb{R}^{|\mathbf{T}|}_{\geq 0} | \bar{\mathbf{R}}\mathbf{x} = \bar{\mathbf{b}} \right\} .$$

Notice that (ii) simply requires us to test that $\bar{\mathbf{b}} \in \mathrm{cone}\left(\bar{\mathbf{R}}\right)$. We can proceed exactly as we did for (i), or use other approaches, to obtain a test statistic $\bar{\xi}\left(\bar{\mathbf{b}}\right)$ such that

$$\bar{\mathbf{b}} \in \mathrm{cone}\left(\bar{\mathbf{R}}\right) \iff \bar{\xi}\left(\bar{\mathbf{b}}\right) \geq 0 ,$$

and then implement the joint test

$$\min\left\{\gamma, \bar{\xi}\left(\bar{\mathbf{b}}\right)\right\} \geq 0 ,$$

using Chernozhukov et al. (2013).

Direct Inference Approach (Preliminary Outline)

Implementing (5) using Chernozhukov et al. (2013) will involve estimating

$$\inf_{\mathbf{v}\in S^{2|\mathbf{T}|-1}} \left\{ \int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(y\mathbf{v}_1 + \mathbf{v}_2;\ \widehat{\mathbf{pf}}_{Y|d,Z}(y) \right) d\mu(y) + h_{\hat{\Lambda}^{(d)}_{\mathbf{T}}(\mathbf{c};\gamma)}(-\mathbf{v}) + \hat{k}_\alpha \cdot \hat{s}(\mathbf{v}) \right\},$$

for an estimated critical value, $\hat{k}_\alpha$, and pointwise standard error, $\hat{s}(\mathbf{v})$. Adding this term, however, generally makes the above a non-convex problem.

- This precision-correction is motivated by the fact that analog estimators for the intersection bounds can be severely biased in finite samples (Manski and Pepper, 2000, 2009)

- When the bounding functions have a "local character" (e.g. density function), standard (e.g., delta-method-based) approaches can fail, motivating the strong approximation based approach of Chernozhukov et al. (2013)

Direct Inference Approach (Preliminary Outline)

While these concerns apply to requirement (i) of our specification test, I don't believe it applies to our proposed test statistic for inference on parameters.

Informally, the idea is that for each $\mathbf{v} \in S^{2|\mathbf{T}|-1}$,

$$\int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(y\mathbf{v}_1 + \mathbf{v}_2;\ \mathbf{pf}_{Y|d,Z}(y)\right)\ d\mu(y) + h_{\Lambda_{\mathbf{T}}^{(d)}(\mathbf{c};\gamma)}(-\mathbf{v})$$

incorporates information from the entire outcome distribution.

This motivates the development of a "direct" approach based on the kernel density estimator, which we sketch below.

Direct Inference Approach (Preliminary Outline)

In order to perform inference based on the estimator

$$\inf_{\mathbf{v}\in S^{2|\mathbf{T}|-1}} \left\{ \int_{\mathcal{Y}} h_{\mathbf{R}_d} \left(y\mathbf{v}_1 + \mathbf{v}_2;\ \hat{\mathbf{pf}}_{Y|d,Z}(y) \right) \, d\mu(y) + h_{\hat{\Lambda}^{(d)}_{\mathbf{T}}(\mathbf{c};\gamma)}(-\mathbf{v}) \right\}, \tag{7}$$

where $\hat{\mathbf{pf}}_{Y|d,Z}$ is a suitable kernel density estimator of $\mathbf{pf}_{Y|d,Z}$, we can proceed by viewing the above estimator as a function of the Kernel CDF (KCDF), defined as the map $y \mapsto \int_{-\infty}^{y} \hat{\mathbf{pf}}_{Y|d,Z}(u)\ d\mu(u)$.

A sufficiently undersmoothed kernel density estimator will produce a consistent KCDF estimator such that the centered and normalized KCDF estimator will converge in law (as a stochastic process) to a generalized Brownian bridge (Theorem 1(iii) in Aït-Sahalia, 1993).

Direct Inference Approach (Informal Outline)

Finally, the proposed approach is:

1. Show that the integrated functional

$$\int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(y\mathbf{v}_1 + \mathbf{v}_2;\ \mathbf{pf}_{Y|d,Z}(y)\right)\ d\mu(y)$$

   is a Hadamard directionally differentiable functional of the CDF, $\int_{-\infty}^{y} \mathbf{pf}_{Y|d,Z}(u)\ d\mu(u)$
2. Then, we have that (7) is Hadamard directionally differentiable as a functional of the function $\int_{\mathcal{Y}} h_{\mathbf{R}_d}\left(y\mathbf{v}_1 + \mathbf{v}_2;\ \widehat{\mathbf{pf}}_{Y|d,Z}(y)\right)\ d\mu(y) + h_{\hat{\Lambda}^{(d)}_{\mathbf{T}}(\mathbf{c};\gamma)}(-\mathbf{v})$
3. The two claims above, along with the results in Fang and Santos (2019) allow us to perform inference on (7) by plugging in bootstrap estimates of the KCDF into an estimator for the directional derivative of (7) (viewed as a functional of the CDF)

Proof of Lemma 2

Proof: By the same argument as Lemma 1, we have that

$$h_{\Theta_{\mathbf{T}|d}(d)}\left(\mathbf{c}_1, \mathbf{c}_2\right) = \int_{\mathcal{Y}} h_{\mathcal{E}_d(y)}\left(\mathbf{c}_1, \mathbf{c}_2\right)\ d\mu\left(y\right)\ .$$

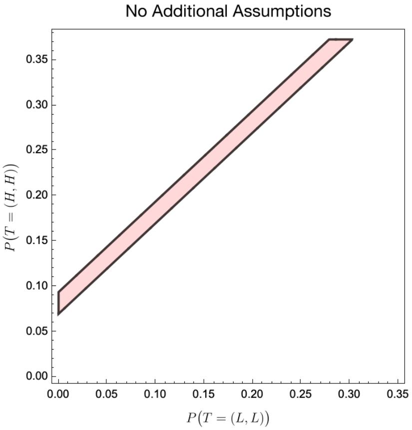
The result now follows by noting that

$$\begin{aligned} h_{\mathcal{E}_d(y)}\left(\mathbf{c}_1, \mathbf{c_2}\right) &:= \sup_{(\mathbf{x}_1, \mathbf{x}_2) \in \mathcal{E}_d(y)} \mathbf{c}_1^\top \mathbf{x}_1 + \mathbf{c}_2^\top \mathbf{x}_2 \\ &= \sup_{(\mathbf{x}_1, \mathbf{x}_2) \in \mathcal{E}_d(y)} \mathbf{c}_1^\top y \mathbf{x}_2 + \mathbf{c}_2^\top \mathbf{x}_2 \\ &= \sup_{\mathbf{x} \in P_{\mathbf{R}_d}\left(\mathbf{pf}_{Y|d,Z}(y)\right)} \left(y\mathbf{c}_1 + \mathbf{c}_2\right)^\top \mathbf{x} \\ &= h_{\mathbf{R}_d}\left(y\mathbf{c}_1 + \mathbf{c}_2;\ \mathbf{pf}_{Y|d,Z}\left(y\right)\right)\ . \end{aligned}$$

Back to Lemma 2

Simulations: Job Training and Wages with Firm Heterogeneity

In this case, the response-type probabilities have 2 degrees of freedom, and can be parameterized by $\Big(\mathbb{P}(T = (L, L)), \mathbb{P}(T = (H, H))\Big)$:

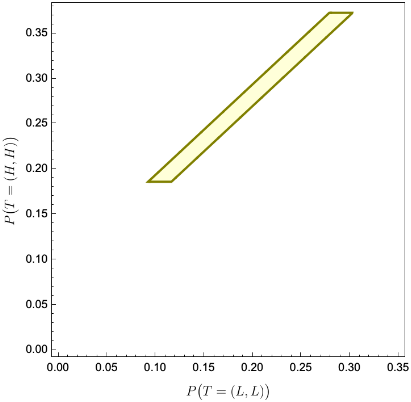


Back to Conclusion

Simulations: Job Training and Wages with Firm Heterogeneity

In this case, the response-type probabilities have 2 degrees of freedom, and can be parameterized by $\left(\mathbb{P}(T=(L,L)), \mathbb{P}(T=(H,H))\right)$:

Assume $P(T=(H,H)) \geq P(T=(H,L))$



Back to Conclusion

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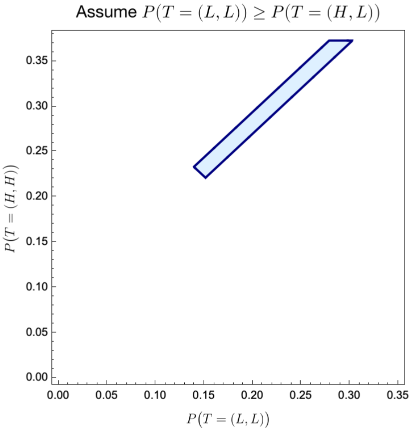
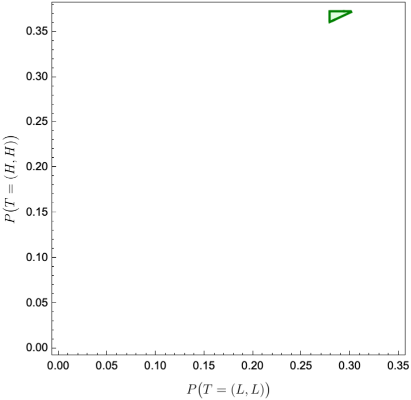


Simulations: Job Training and Wages with Firm Heterogeneity

In this case, the response-type probabilities have 2 degrees of freedom, and can be parameterized by $\left(\mathbb{P}(T=(L,L)),\mathbb{P}(T=(H,H))\right)$:

Assume $\min_{\tau} P(T=\tau) = P(T=(H,L))$



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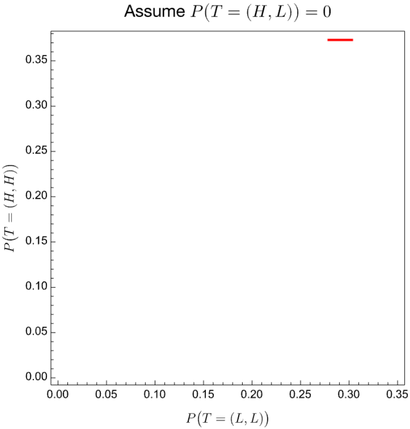
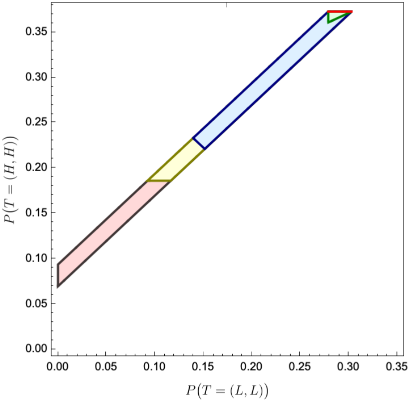
Simulations: Job Training and Wages with Firm Heterogeneity

In this case, the response-type probabilities have 2 degrees of freedom, and can be parameterized by $\left(\mathbb{P}(T = (L,L)), \mathbb{P}(T = (H,H))\right)$:



Simulations: Job Training and Wages with Firm Heterogeneity

In this case, the response-type probabilities have 2 degrees of freedom, and can be parameterized by $\Big(\mathbb{P}(T=(L,L)),\mathbb{P}(T=(H,H))\Big)$:



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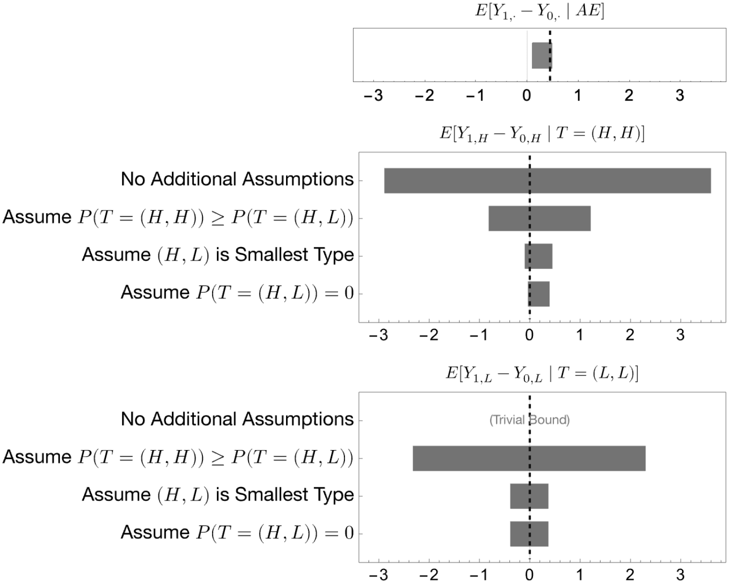
Simulation 1: Job Training and Wages with Firm Heterogeneity



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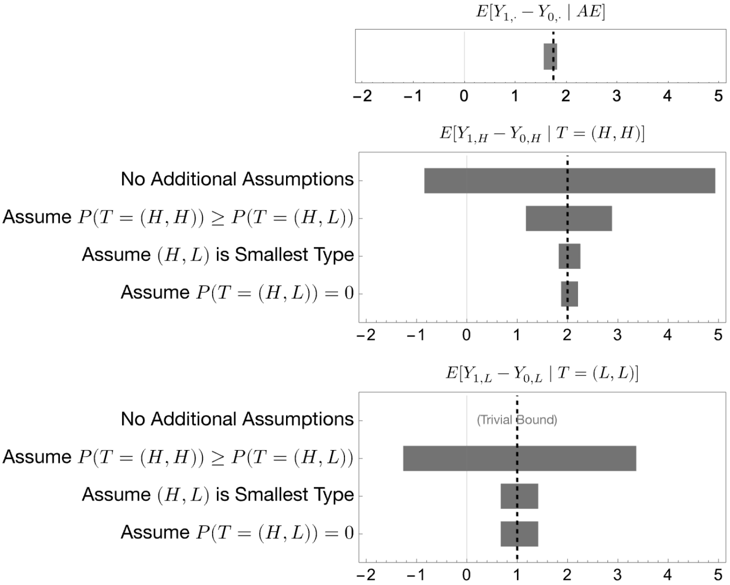
Simulation 2: Job Training and Wages with Firm Heterogeneity



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